CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2014

CL 308 Geotechnical Engineering-II

Date: 12.05.2014, Monday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) List out the types of borings for soil exploration. [02]

(b) Explain in detail the standard penetration test to calculate N-value. [05]

Q - 2 (a) Enlist the different forces acting on the well foundation. Explain any three in detail. [07]

(b) Give the detail note on grip length. [03]

OR

Q - 2 (a) Draw different shapes of wells. Mention in which conditions they are favorable? [05]

(b) Explain well foundation in detail & show all parts of it with neat sketch. [05]

(c) Define lateral earth pressure. Discuss *any three* cases of active earth pressure with formulae. [04]

Q - 3 (a) A retaining wall, 5 m high supports a backfill $C = 20kN/m^2$, $\emptyset = 30^\circ$, $\gamma = 20kN/m^2$, with horizontal top, flush with the top of wall. The backfill carries a surface load of $25kN/m^2$. If the wall is pushed towards the backfill, compute the total passive pressure on the wall, & its point of application. [05]

OR

Q - 3 (a) For the retaining wall shown in **figure-1**, assume the active condition. Use Rankine's active earth pressure theory to determine; [05]

(i) Active force per meter of wall, and

(ii) The location of the resultant line of action ($C = 0$).

(b) A raft of size 4mX4m carries a load of $200kN/m^2$. Determine vertical stress increment at a point 4m below the centre of the loaded area using Boussinesq's theory. Compare the results with that obtained by equivalent point load method (for 1 equivalent point load & 4 equivalent point loads). [05]

(c) Discuss the basis of construction of Newmark's Influence chart and its use. [04]

OR

(c) Enlist the assumptions made for Boussinesq's theory. Give the equation for vertical pressure under a uniformly loaded circular area. [04]

SECTION – II

Q - 4 (a) Explain in detail the method to locate the centre of critical failure surface. [05]

(b) List various methods of improving stability of slopes. [02]

Q - 5 (a) A square column foundation is to be designed for a gross allowable total load of 250 kN. If the load is inclined at an angle of 15° to the vertical, determine the width of the foundation. Take F.O.S. = 3 and use Vesic's equation. $\phi = 35°$, $c' = 5 kN/m^2$, $\gamma = 19 kN/m^3$ and depth of foundation = 1 m. Use **table-1** for bearing capacity factors and **table-2** for formulae to calculate different factors. [07]

(b) Write assumptions made in the theory of Terzaghi's analysis. Give bearing capacity equations for strip, circular and square footing as per Terzaghi's theory. [05]

OR

Q - 5 (a) A square footing has dimensions of 2mX2m and a depth of 2m. Determine its ultimate bearing capacity in pure clay with an unconfined strength of $0.15 N/mm^2$, $\phi = 0°$ and $\gamma = 1.7 gm/cm^3$. Assume Terzaghi's factors as $N_c = 5.7$; $N_q = 1.4$; $N_\gamma = 0$. [07]

(b) (i) Differentiate between general shear failure and local shear failure. [02]

(ii) Define Any Three; [03]

Bearing capacity, Ultimate bearing capacity, Footing and Net safe-bearing capacity.

(c) Differentiate between end bearing pile and friction pile. [02]

Q - 6 (a) An embankment is constructed by sandy clay having cohesion of $30 kN/m^2$, angle of internal friction is 20° and unit weight of $20 kN/m^3$. The height of embankment is 10m, $\alpha=25°$ & $\beta=15°$. Determine the factor of safety using method of slices, by considering toe failure. [05]

(b) Write detail short note on group action of pile with neat sketch. [03]

(c) A group of 9 piles, 9 m long is used as the foundation of column. The piles are 30 cm in diameter with centre to centre spacing 90 cm. the subsoil consists of clay with cohesion $85 kN/m^2$. Estimate safe load. Take F.O.S. = 3. [06]

OR

(b) Define negative skin friction. How it affects the pile load carrying capacity? [03]

(c) Explain pile load test with neat experimental setup sketch. [06]

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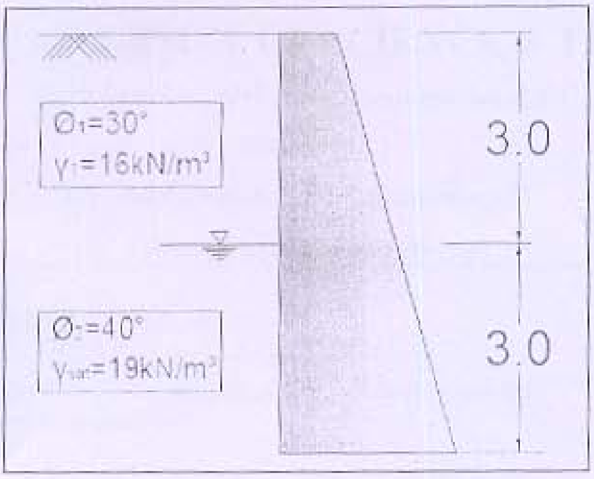

Figure-1

Table-1

TABLE 1 BEARING CAPACITY FACTORS

(*Clause* 5.1.1)

ϕ (Degrees)	Bearing Capacity Factors N_c	N_q	N_γ
0	5·14	1·00	0·00
5	6·49	1·57	0·45
10	8·35	2·47	1·22
15	10·98	3·94	2·65
20	14·83	6·40	5·39
25	20·72	10·66	10·88
30	30·14	18·40	22·40
35	46·12	33·30	48·03
40	75·31	64·20	109·41
45	138·88	134·88	271·76
50	266·89	319·07	762·89

Table-2 Formulae to calculate shape, depth & inclination factors.

Shape Factors	*Depth Factors*	*Inclination Factors*
$Sc = 1 + (Nq / Nc)$	$dc = 1 + 0.4\,(Df/B)$	$ic = iq = (1-\alpha/90)^2$
$Sq = 1 + \tan Ø'$	$dq = 1 + 2\tan Ø'(1-\sin Ø')^2(Df/B)$	$i\gamma = (1-\alpha/Ø')^2$
$S\gamma = 0.60$	$d\gamma = 1.00$	